Building a Contract Dataset with a Fine-Tuned Language Model, and What a Commercial Database Measures

Yangen Li*
University of Iowa

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Abstract

I fine-tune an open-weight 14-billion-parameter language model on 20,000 machine-labeled SEC exhibits and use it to read the EDGAR exhibit record from 2005 through June 2026. That is 1.38 million documents, classified on three consumer GPUs. What fine-tuning buys is precision. On held-out documents the tuned model reaches an F_1 of 0.978, against 0.919 for the best prompt I found, and it makes 27 false positives where prompting makes 140. But a classifier that reads documents well does not by itself give a correct dataset. I propose a way to check one: draw stratified samples, judge them against an independent source, read every case twice, and plant control cases whose answer is known. On 410 cases, almost none of the remaining error came from misreading documents. It came from the code that decides which person and which year a document belongs to. The same check, applied to the leading commercial database, shows that its employment-agreement flag and mine do not measure the same thing. Among CEOs it never credits with an agreement, 21.8 percent have a formal employment agreement on file. And when the document is an offer letter, the vendor calls it an agreement only 57.7 percent of the time. Firms must file an executed contract, but they choose how to describe it. A measure built from filings sees the document. A measure built from proxy statements sees the description.

*Code for every stage of the pipeline is public at `https://github.com/yangenli/llm-finetuning`.

1 Introduction

Contracts are the main evidence in the economics of organizations, and they are among the least used. The obstacle is collection. Gillan et al. [2009] read proxy statements and SEC filings by hand for the 494 US firms in the S&P 500 as of January 2000. They report explicit employment agreements for 45.6 percent of them. The work is exemplary, and it does not scale. Commercial vendors sell coverage at scale but not construction: the researcher gets a flag and cannot inspect what produced it. Between the two sits the entire filed record, every material contract a US registrant has been required to file. It has gone unread, because reading it needed either a decade of research assistants or a machine that could understand legal prose.

This paper does three things. First, it gives a working recipe for the third option: a fine-tuned open-weight language model that classifies 1.38 million SEC exhibits on hardware one researcher can own. Second, it proposes a way to validate a dataset built like this. The field has no settled answer to "how do you know the machine got it right?", and classifier accuracy on a held-out split is not that answer. Third, it reports what the validation found, which turned out to be more interesting than the recipe.

Three results are worth stating up front.

Fine-tuning buys precision, and precision is what a dataset needs. A 14B model prompted zero-shot reaches an F_1 of 0.892 on this task. The same model with eight in-context examples reaches 0.919. Fine-tuned on 20,000 labeled heads, it reaches 0.978. The gap in F_1 is modest. Its composition is the point. Prompting matches the tuned model's recall almost exactly, and pays for it with five times the false positives. It also costs five times the compute at inference, because eight exemplars need about 9,000 prompt tokens per document against 1,900 for the tuned model. Prompting instead of tuning saves nothing here. It buys a dirtier dataset and a slower run.

Model accuracy is not dataset accuracy. My classifier reads documents almost perfectly. The dataset built from it was wrong far more often than that. The error was downstream, in three ordinary decisions: whether a document is dated by its signature or by its filing, whether a contract signed while the person was chief operating officer counts once they become chief executive, and whether a truncated name matches a roster entry. None of these is visible to a held-out evaluation of the model. Together they are this paper's central methodological claim.

The comparison with a commercial database revealed a construct problem, not a quality problem. I expected to find that the vendor made mistakes. It does. But its larger deviation from my measure is definitional and systematic. Firms differ in whether they call an executed offer letter an employment agreement. A vendor that reads proxy statements inherits that choice. A measure that reads filings does not.

Relation to prior work

This paper sits between three literatures. The first reads executive contracts directly. Gillan et al. [2009] hand-collect employment agreements for the S&P 500. Schwab and Thomas [2006] document what executives bargain for. Song and Wan [2017] relate explicit contracting to pay. That work establishes the object and its economic interest. What it cannot do is cover the filed record, which is why the variable is usually bought rather than built.

The second literature treats a disclosure as a choice rather than a fact. Verrecchia and Weber [2006] on redaction is the closest case: what the researcher observes is produced jointly by the event and by the firm's decision about how to describe it. My vendor result applies that idea to a control variable. An employment-agreement flag read out of proxy language inherits the firm's vocabulary. One read out of the exhibit record does not.

The third literature is the new practice of building research datasets with language models, where reporting conventions are not yet settled. My contribution there is procedural. A held-out F_1 is not evidence about a dataset, and I give the validation design I think should accompany one.

Roadmap

Section 2 describes the pipeline: corpus, task, labels, fine-tuning and serving. Section 3 evaluates the model against the held-out labels and against prompting. Section 4 gives the validation design, and reports what adjudicating 410 cases found about my own dataset. Section 5 applies the same machinery to the commercial measure. Sections 6 and 7 discuss and conclude. Three appendices hold material that would interrupt the argument: a comparison I withdrew and replaced (Appendix A), the de-anchored and adversarial recheck of the adjudication (Appendix B), and a defect in the released adapter (Appendix C).

Code for every stage is public. This covers crawling EDGAR, extracting and deduplicating exhibits, drawing and labeling the training sample, the fine-tune, the serving stack, and the matching, dating and adjudication scripts behind Sections 4 and 5. It is at `https://github.com/yangenli/llm-finetuning`.

2 The Pipeline

2.1 Corpus

The unit is a filed exhibit. I crawl the EDGAR submission record and extract every EX-10 and EX-99 document. Identical attachments filed repeatedly across years are deduplicated.

The classification frame is 1,382,023 documents dated from 2005 through June 2026. Restricted to fiscal years 2005–2024, the window all later tables use, it is 1,296,936.

The two arms are not symmetric. All 1,212,928 EX-10 documents go to the model with no filter. Employment agreements are an EX-10 genre, and a prefilter would trade recall for compute I did not need to save. The EX-99 arm is gated by keyword to 169,095 documents. That was a cost decision, and I audited it. Of 1,113 documents drawn at random from the excluded remainder, none was an executive contract. This puts an upper bound of 0.3 percent at 95 percent confidence. The remainder is large enough that even 0.3 percent would be a material number of documents.

2.2 The task and the schema

The model reads the first 4,000 characters of a tag-stripped document. This is the head, where a contract states what it is and who signed it. It emits one JSON object:

```
{"contract": 0|1, "types": [...], "person": str|null, "title": str|null,
"ceo": 0|1, "executed": 0|1, "amendment": 0|1}
```

`types` draws on a 21-item vocabulary. `contract` covers a broad executive-compensation class, of which employment agreements are one genre. The CEO employment dataset is a filtered subset of it, and Section 4.5 gives the funnel.

Two fields do more work than their size suggests. `executed` separates signed instruments from the blank template agreements registrants also file. Those carry `[NAME]` placeholders and would otherwise enter the dataset as contracts. `person` is the join key to any executive roster. It is where the pipeline's residual error concentrates (Section 4.3).

One prompt file holds the labeling instructions, the fine-tuning target format and the inference prompt, so that all three are defined in one place. I recommend this, with one condition: the file must be pinned and versioned with the checkpoint. Mine was corrected after the training run, and the shipped adapter had already learned a defect from the earlier version (Appendix C).

2.3 Labels

Training labels come from 20,000 exhibit heads, drawn in three strata:

- 14,457 documents that a 21-column keyword classifier flagged as candidate contracts. It reads the description, the filename and the document head.
- 4,430 EX-10 documents that the keyword classifier rejected.
- 1,113 EX-99 rejects.

Sampling the rejects is what lets the false-negative rate of the keyword screen be estimated rather than assumed. It is also why the EX-10 arm ends up bypassing the screen entirely. Of the 20,000 documents, 8,190 are labeled as contracts and 477 as CEO-level.

2.4 Fine-tuning

I fine-tune Qwen3-14B with QLoRA, starting from a pre-quantized 4-bit checkpoint. The settings are rank 16, $\alpha = 32$, dropout 0.05, and adapters on all seven projection matrices. The learning rate is 2×10^{-4} with cosine decay and 5 percent warmup, over two epochs, at batch size 4 with gradient accumulation 2. The maximum sequence length is 2,304 tokens. Quantization is NF4 double quantization with bfloat16 compute, and the optimizer is 8-bit AdamW. The split is 18,000 training and 2,000 test, stratified, seed 42. The run environment is a single RTX 5090.

2.5 Serving

Production inference does not use the fine-tuned weights directly. I convert the base model to a 6-bit GGUF quantization (12.1 GB) and apply the LoRA adapter (128 MB) at run time under `llama.cpp`. One base file can then serve several tasks, and the whole model stays resident on a consumer card. Section 3.2 shows that this is also the most accurate of the served configurations, not just the most convenient. In every served configuration the "base" is the pre-quantized 4-bit release, dequantized and re-quantized to 6 bits. It is not the vendor's original bfloat16 weights.

Three GPUs run disjoint year ranges at the same time, with greedy decoding and prompt caching. Each document occupies a 3,072-token slot. The RTX 5090 served 24 slots at a 73,728-token context, the 4090 16 slots at 49,152, and the 5080 4 slots at 12,288.

Throughput comes from a two-pass schedule. Most exhibits are not employment contracts, and the first field of the output object is `contract`. Pass one therefore generates eight tokens per document, just enough to read `{"contract": 0` or `1`, and stops. Pass two generates the full 160-token object, and only for the documents that survived pass one.

This cuts decoding on most documents by roughly a factor of twenty. The wall-clock gain is smaller, because with a prompt of about 1,900 tokens prefill dominates. I report the arithmetic rather than a measured end-to-end speedup.

The measured rate is 1.6 to 1.9 documents per second per GPU. That puts the full frame at roughly 210 to 250 GPU-hours.

3 How Accurate Is the Classifier?

3.1 Fine-tuning against prompting

Table 1 evaluates four configurations on the same 2,000 held-out documents. The comparison that matters is the first two rows: the same model, tuned and untuned.

Table 1: Agreement with the held-out label set, 2,000 documents

Configuration	Precision	Recall	F_1	Accuracy	Person exact	False positives	Prompt tokens
Fine-tuned (QLoRA)	**0.968**	0.988	**0.978**	**0.981**	0.979	**27**	1,900
Zero-shot, same base model	0.906	0.879	0.892	0.912	0.881	76	1,677
Eight-shot, 4 of 8 positive	0.851	**0.994**	0.917	0.925	0.919	144	9,097
(across three draws)	(0.014)		(0.007)				
Eight-shot, 6 of 8 positive	0.855	**0.994**	0.919	0.927	0.922	140	8,945
(across three draws)	(0.027)		(0.014)				

Notes. Binary contract classification on the same 2,000 held-out documents (829 labelled contracts, 1,167 not). The few-shot rows are means over three independent exemplar draws, with the standard deviation across draws beneath. Exemplars come from the 18,000-document training split only; every test document key was excluded. All rows are served through the production stack, which for row one means the 6-bit base with a runtime adapter. Evaluating the same adapter unquantized and in process gives $P = 0.970$, $R = 0.986$, $F_1 = 0.978$. Row one scores 1,985 parseable outputs; the other rows score 1,996. "Person exact" is exact-string agreement on the extracted counterparty name. It is computed only where both the label and the prediction are contracts, so its denominator differs by row.

Precision and recall tell different stories here. Fine-tuning does not find more contracts than prompting with exemplars. It recovers 819 of the 829, where eight-shot prompting recovers 821. What it does is reach that recall without dragging in the rest of the corpus. The prompted model pays for its 0.994 recall with 140 false positives. The tuned model reaches 0.988 with 27. Zero-shot prompting sits at the other end of the same trade. It buys precision of 0.906 by missing 100 real contracts.

The two prompted configurations are not worse versions of the tuned model. They are different points on roughly one curve. Exemplars push the model toward saying yes, which raises recall and lowers precision together. Fine-tuning moves the curve instead of the point.

It beats zero-shot on both axes. Against the best prompted configuration I found, it gives up 0.006 of recall and gains 0.113 of precision. On 1.38 million documents, that is the difference between a dataset you can use and one you have to read again.

An earlier version of this table reported a much larger gap, from two uncontrolled single-draw runs. I withdraw those numbers. Appendix A documents the replacement, and what it teaches about designing a comparison of this kind.

3.2 Serving configurations

Table 2: Served configurations, 600-document evaluation

Served model	F_1	Accuracy	Docs/s
6-bit base + runtime adapter (production)	**0.980**	**0.983**	1.17
6-bit merged fine-tuned model	0.948	0.956	1.72
4-bit merged fine-tuned model	0.945	0.954	1.93
6-bit base, no adapter, zero-shot	0.897	0.915	3.53

Notes. At identical quantization, the adapter is worth 8.3 points of F_1: row one against row four. This replicates Table 1 on the production hardware. Applying the adapter at run time also beats merging it into the weights before quantization, by about three points. I did not expect this, and report it as a practitioner finding. Throughput here is single-pass full generation on a fixed evaluation set at client concurrency 32. It is not the two-pass production path, so it is not comparable to the 1.6 to 1.9 documents per second of Section 2.5.

4 The Dataset Is Less Accurate Than the Classifier

A model that reproduces 98 percent of labels does not give a dataset that is 98 percent correct. A dataset is not a pile of documents. It is a set of claims about people and periods: this CEO, at this firm, in this year, had an agreement. Every such claim requires attributing documents to people and to years. Those attribution rules are ordinary code. They are invisible to any held-out evaluation of the model. In my case they held nearly all the error.

4.1 A validation design for machine-built datasets

I propose the following, and use it here.

Adjudicate against an independent source, not against the model. Agreement between two model passes measures consistency, not accuracy. My independent source is the proxy statement. It discusses the same arrangements under a disclosure mandate, and it is not derived from the exhibits.

Stratify on the disagreement structure. With two measures there are four cells. Sampling only where they disagree gives conditional error rates and no unconditional one. Sampling at random spends most of the budget where both agree. I sample all four cells and reweight by cell size.

Package the evidence, and be precise about what is hidden. Each case becomes a packet. It holds the company, the executive, the fiscal year end, every employment or offer-letter exhibit at that firm matching the executive's surname, the signing and filing dates, the role recorded on the document, and a keyword-windowed excerpt of that year's proxy. The packet never reveals the vendor's answer. It does not conceal mine. A packet that reports no matching exhibit implies that my measure recorded none, and each packet prints the classifier's own type label, extracted party and match tier above the questions the rater is asked. In round one, all 64 packets reading "none found" are cases where my measure is zero.

Plant controls and read twice. Twenty cases on which both measures agree were mixed in unlabeled. The both-say-yes controls returned a clean yes 10 times out of 10. The both-say-no controls returned a clean no only 4 times out of 10, with 5 unresolved and 2 positive. Establishing absence from a keyword-windowed proxy excerpt is genuinely hard.

Say who the adjudicators are. Mine are language-model agents following a fixed rubric, not human coders. Read-to-read agreement on the existence question was 97.4 percent ($n = 151$) and 97.2 percent ($n = 141$) on the two double-read subsets, which oversample unsettled cases by design. Agreement on the other recorded fields is lower: 92.1 percent for whether a formal agreement was in force, 87.4 percent for how the proxy described the instrument, and 82.5 percent for the role under which it was signed.

Restrict to cases the source can settle. Foreign private issuers file no proxy statement, and are exempt from individual-executive compensation disclosure. They resolve at 46 percent, against 95 percent for US registrants, and they make up roughly 30 percent of the cell where both measures record nothing. For such a firm a zero means "not observable under its filing regime", not "no contract". They belong outside the frame, not inside it as uncertainty. Restricting to US registrants roughly halves the unresolved share, from 10.3 to 4.8 percent.

4.2 Where the errors came from

The model was rarely at fault. Adjudicators inspected 398 exhibits across the two rounds. They found one document matched to the wrong person and two typed incorrectly. The two mistypes were a technical services agreement matched to an energy-company CEO, and a pair of letters from a dissident stockholder in a proxy contest. The wrong-person case was not a model error. The model extracted the name correctly. My matcher then attached it to a different executive with the same surname.

The dataset missed 87 contracts:

- 37 (43 percent): I hold the contract, but dated it by its filing date.
- 28 (32 percent): the instrument was signed in a prior role.
- 11: only a later or amended-and-restated version was filed. Its recitals name the original, which I do not parse.
- 8: no version was ever filed.
- 3: the classifier erred.

Two of these problems will appear in any dataset built from filings.

Filing dates are not signing dates. I parse the execution date from the document itself, which works for 83.8 percent of them, and use the filing date otherwise. The median gap between the two is 50 days. For 41 percent of documents it exceeds 90 days, and for 24 percent it exceeds a year. The gap follows the disclosure rules. An agreement attached to an appointment 8-K arrives in a median of 5 days. One swept into the next periodic report takes about 125 days. One first disclosed in an IPO registration statement takes 413 days. Contracts signed while the firm was private, or before the signer was a named executive officer, surface years later. Dating by filing understates the contract rate by 6.1 percentage points in a CEO's first year, and by about one point after that.

People change roles; documents do not. A letter can fix a chief operating officer's terms when they are promoted to chief executive. It is filed as a COO document, so a CEO-title filter drops it, even though it governs the CEO's employment. Counting instruments signed in a prior role recovers a third of the remaining misses. It also needs a limit: an old agreement from a prior job does not make its holder a contracted CEO.

4.3 Matching names to executives

The counterparty name links a document to a roster. It fails in ways that an audit is needed to see:

- Diacritics stripped upstream truncate surnames: "Carol Tom" for Carol Tomé.
- Compound surnames arrive split: "Bl zquez", "Del Rio".
- Some documents carry a first name or a nickname only: "Bob" at Intel, "Marvin" at Lowe's.

- Rosters lead with an initial where documents use the middle name: "WENGER, E. PHILIP" is Philip Wenger.
- 4,449 of the 143,618 primary employment and offer-letter instruments (3.1 percent) name nobody. Of those, 65.5 percent carry blank-template markers and 85.7 percent are flagged unexecuted, against 0.1 percent of named documents.

I rebuilt the matcher as a tiered rule: exact, nickname, initial, fuzzy surname, first-name-only behind a uniqueness guard, and surname-only. Any table can then be recomputed at a stricter tier, and the loose tiers can be audited.

The adjudication identified 22 matching failures, of which 18 are recoverable in principle. The new matcher recovers 17. It loses none of 300 positive controls, and it reduces false positives on a 50-case true-negative set from 8 to 7, at the tiers used for the headline tables. The process is worth reporting alongside the result. Two earlier versions passed the recovery test while silently losing matches elsewhere. They would have shipped if I had not tested both directions at once.

4.4 Rechecking the adjudication

The packets in Section 4.1 conceal the vendor's answer but not mine. And the distinction between a formal agreement and a letter, which carries this paper's headline numbers, had never been checked against anything. I therefore rebuilt the packets for the 189 US-registrant cases in the two cells that carry those numbers. The rebuilt packets remove the classifier's type label, its extracted counterparty and the match tier. I then read the cases twice more: once blind to my own measure, and once with a rater instructed to refute the resulting verdicts. Appendix B reports the exercise in full. Three results carry into the argument.

Anchoring changed almost nothing. Across the 183 cases both passes resolved, the de-anchored verdict agrees with the anchored one 99.5 percent of the time. One verdict in 189 moved.

The instrument-type split is not an artifact of the classifier's own labels. Typing 168 documents from their text alone, the de-anchored rater reads a formal agreement in 102 and a letter in 57. Where it reads a formal agreement, the pipeline recorded one in force in 81 of 99 resolved cases. Where it reads a letter, the pipeline recorded one in 3 of 54.

The existence claims survive attack. The adversary refutes 3 of 188. And 144 of the 147 cases behind the vendor comparison (98.0 percent) are confirmed by all three passes.

4.5 From documents to observations

The funnel, for scale:

- 1,382,023 documents classified.
- 517,964 with `contract` = 1.
- 139,719 executed primary employment or offer-letter instruments. Of these, 112,332 are formal employment agreements and 27,387 are offer or letter agreements.
- 30,678 matched to a CEO.
- 22,306 CEO-firm spells in the US-registrant panel.

5 Comparison with a Commercial Database

Having validated my own measure, I compare it with Equilar's employment-agreement flag. That flag is built from proxy statements, and is a standard source for this variable.

The unit is the CEO-firm spell. I require the vendor to record no agreement in *any* year of a CEO's tenure, so that a contradiction cannot be a dispute about start dates. The headline comparisons cover spells beginning in 2013 or later. I report the full 2005–2024 panel for reference.

The restriction is necessary, because the vendor's series contains a coding-regime break. The share of CEO-years flagged yes falls from 80 to 82 percent through 2009, down to 33 to 38 percent from 2012. Of CEOs observed at the same firm in both 2009 and 2012, 59.5 percent flip to no. Expiry explains part of this, and cannot explain all of it: 36.1 percent of CEOs who signed a fresh formal agreement in 2010 to 2012 also flip. The pre-2010 level also sits far above the 45.6 percent that Gillan et al. [2009] hand-collect for S&P 500 firms in January 2000. My own cohort gives about 41 percent for 2005.

5.1 Where the two measures disagree

Table 3: What I hold for CEOs the vendor never credits with an agreement

	CEOs	Any instrument	Offer letter only	**Formal agreement**
All spells, 2005–2024	7,860	27.0%	8.5%	**18.4%**
Spells beginning 2013 or later	5,087	32.9%	11.2%	**21.8%**
and tenure $\geq$ 3 years	2,307	35.0%	12.2%	**22.8%**

Notes. US registrants. The vendor records no agreement in any year of the spell. An instrument counts if it is dated on or before the end of the spell, using the parsed execution date where available and the filing date otherwise. The execution date parses for 18.4 percent of instruments in this frame, so the count is conservative. Restricting the 2013+ row to original agreements, excluding amendments, gives 16.6 percent. Dropping matches that rest on a surname or a first name alone gives 31.7 percent in the first column; dropping surname-only matches alone gives 32.2 percent. This table incorporates execution-date dating.

Adjudication of the same cell confirms the direction, and separates two questions. Among 147 adjudicated US-registrant cases where I hold an instrument and the vendor records none, a written instrument was found in 145 of 146 resolved cases. Asked instead whether a *formal* employment agreement was in force, raters said yes in 79 of 141. Reporting only the first number would overstate the vendor's error rate by about a factor of two. The gap between the two numbers is the offer-letter question.

5.2 Firms describe the same document differently

If the vendor applied a consistent rule to offer letters, then among CEOs whose only instrument is a letter its flag would be near zero or near one. It is neither.

Table 4: Does the vendor record an agreement, given the instrument I hold?

What I hold	CEOs	Vendor records an agreement
Formal employment agreement	4,643	76.2%
Offer letter only	**1,344**	**57.7%**
Nothing I can see	4,357	21.7%

Notes. CEO-firm spells beginning 2013 or later, US registrants. Within the offer-letter row, quartiles computed on the 1,286 spells with non-missing market capitalization give 60.7, 66.9, 58.9 and 42.5 percent, from the smallest to the largest. The series is not monotone, since its maximum is in the second quartile. I therefore describe it as a large-cap discontinuity rather than a gradient.

The 57.7 percent is the finding. The same kind of document is credited at one firm and denied at another, and denial concentrates among the largest firms. Two cases make the mechanism concrete. Both verify against the exhibits.

- **Microsoft.** Satya Nadella's instrument is a letter dated 3 February 2014: "Dear Satya: On behalf of the Board of Directors, I am pleased to offer you the position of Chief Executive Officer". It fixes salary, incentive targets and long-term awards. Microsoft's proxy lists "No employment agreements" among its governance highlights, and provides severance through a plan open to designated executives generally. The vendor records no agreement for eleven years, and on its own terms is right.
- **Visa.** Alfred Kelly's instrument is also an offer letter. Visa's proxy devotes a section to it: "We have outstanding obligations under an executed offer letter with Mr. Kelly". The proxy separately notes the absence of *fixed-term* agreements. The vendor records an agreement.

Adjudicated cases show the vendor tracking the description as much as the document. Among 299 US-registrant cases where an instrument was found to exist, the vendor's yes-rate is:

- 0.0 percent where the proxy states that the firm has no agreements ($n = 9$).
- 25.0 percent where the proxy describes an arrangement with no written instrument ($n = 8$).
- 46.0 percent where the proxy calls the instrument a letter ($n = 50$).
- 43.4 percent where the proxy calls it an employment agreement ($n = 182$).
- 67.6 percent where the proxy is silent ($n = 37$).

The two middle categories are statistically indistinguishable. That is the point. What the proxy calls the instrument does not move the flag nearly as much as whether the proxy denies having one, which drives the flag to zero. The proxy's own label is accurate about the document. Raters found a formal agreement in force in 2 of 51 resolved cases the proxy called a letter, against 160 of 180 it called an agreement. So firms are not mislabeling. They are choosing different words, and different arrangements, and the vendor inherits both.

6 Discussion

For researchers building datasets with language models, the results here suggest a reordering of effort. The model is the part most likely to be tuned, benchmarked and reported. In my case it was the part least likely to be wrong. Held-out classification metrics measure the easiest link in the chain. The rules that turn documents into observations are dating, entity resolution, and role and period attribution. They are undocumented in most papers, and they held nearly all of my error. I would encourage authors to report how documents are dated and matched alongside the classifier metrics, and reviewers to ask for it.

For researchers using contract data, the vendor comparison raises a question that is rarely asked of a control variable: whose description is this? A proxy-derived flag will correlate with governance quality, institutional ownership and proxy-advisor pressure partly through language, because those forces plausibly shape how a firm describes an arrangement as well as which arrangement it enters. A filing-derived measure's errors do not depend on the firm's disclosure vocabulary.

Cost matters here, because it decides who can do this work. The classification behind this dataset ran on three consumer graphics cards in a few hundred GPU-hours, at the marginal cost of electricity. Full-text second-pass extraction through a frontier API is a different order of expense. I measured it at roughly 600,000 tokens per document on a related task. Head classification of a 4,000-character input is not the same job, and the two should not be compared directly.

Within the classification step the comparison is clean, and it runs the same way on both axes. A tuned model answers from a 1,900-token prompt, where an eight-shot prompt costs about 9,000. The tuning is therefore repaid in inference compute on a corpus this size, whatever the accuracy. And it is also the more accurate of the two, by 27 false positives against 140. The usual reason to prompt rather than tune is that tuning costs a day of work up front. That day buys nothing back here.

One last point, about where the money should go. Of the 87 contracts this dataset missed, the classifier misread 3. Eight were never filed by anyone. The remaining 76 were lost to the dating rule, to the role filter, or to a successor document I do not parse. That is code which never looks at what the document says. A better reader would recover the 3 and leave the 76 where they are. So the case for a cheap local model is not only that it is cheap. It is that the expensive alternative would have bought almost nothing here, and that the effort it saves is better spent on the rules in Section 4.

7 Conclusion

A single researcher can now read the filed record. The recipe is a fine-tuned open-weight model on consumer hardware. The part worth copying is not the architecture but the accounting. Fine-tuning bought precision rather than recall, and precision is what a dataset needs, because a false positive enters the data and stays there.

The more durable lesson is about where such a dataset goes wrong. Mine read documents almost perfectly, and was still wrong more often than its F_1 implied. The error was in dating, in entity resolution and in role attribution. That is ordinary code, and no held-out evaluation can see it. Validating against an independent source, rather than against the model, is what surfaced it.

The same design turned an expected quality comparison with a commercial vendor into a comparison of constructs. The vendor is not careless. It reads what firms say about their arrangements, and I read the arrangements. The two diverge exactly where firms differ in what they call the same document. A researcher choosing between the two measures is choosing between the instrument and its description, and should say which one the hypothesis is about.

A The withdrawn few-shot comparison

An earlier version of Table 1 reported an eight-shot F_1 of 0.737, with precision collapsing to 0.584 and 589 false positives. I withdraw those numbers. They came from two single-draw runs whose exemplar-selection script was not retained. The two conditions differed both in the positive ratio and in which documents were drawn, so nothing could be attributed to either. Redone under the design of Section 3.1, the same nominal condition gives an F_1 of 0.917 with 144 false positives. Prompting does not collapse on this task.

Three things support the replacement.

First, an internal check. Re-run through the new harness, zero-shot reproduces the earlier figure to three decimals: F_1 0.892 against 0.894, and precision 0.906 against 0.909. The harness is therefore equivalent, and the divergence is in the exemplars rather than in the measurement.

Second, the new exemplars are shown at their full 4,000-character length, where the earlier ones were truncated. This makes the new run a better-specified version of the same idea, not a replication of it.

Third, and most usefully for anyone designing such a comparison, the draw dominates the design choice. Across the ratios I tried, condition means run from 0.900 to 0.920. Within a single condition, three draws of eight exemplars span 0.839 to 0.943. A comparison of two single-draw conditions is a comparison of noise, which is what the withdrawn rows were. I report standard deviations across draws for this reason. Reading a single seed as a result mid-run forced a retraction when the next seed returned the best score of the sweep.

The per-stratum cut survives the correction in direction, if not in size. On documents the keyword screen had already rejected, which are the easiest negatives, the zero-shot model has a false-positive rate of 4.1 percent and the prompted model 5.3 to 10.1 percent. Exemplars do make the model more willing to call an unrelated document a contract. They do not make it useless.

B The de-anchored, adversarial replication

The evidence packets of Section 4.1 conceal the vendor's answer but leak my own. And the distinction between a formal agreement and a letter, which carries this paper's headline numbers, had never been checked against anything. I therefore rebuilt the packets for the 189 US-registrant cases in the two cells that carry those numbers, and read them twice more. Section 4.4 states the result. This appendix reports it in full.

The rebuilt packets remove the classifier's type label, its extracted counterparty and the match tier. They also reproduce each exhibit's own opening text, properly stripped of markup. The rater must therefore identify the parties and type the instrument from the document, rather than confirm a label already supplied. A second pass then gives an adversarial rater the packet and the resulting verdicts, with instructions to refute them and to default to "refuted" when unsatisfied.

Anchoring changed almost nothing. Across the 183 cases both passes resolved, the de-anchored verdict agrees with the anchored one 99.5 percent of the time. A single verdict moved, from yes to no. The unresolved share *fell* from 3.2 to 1.6 percent, because the rebuilt packets fixed a defect in the original build that pasted some exhibits in as raw markup. The leak was a real design fault, and its measured consequence was one case in 189.

The instrument-type split holds. Typing 168 documents from their text alone, the de-anchored rater calls 102 formal employment agreements, 57 offer or letter agreements, 4 amendments only, and 5 not employment instruments at all. Where it reads a formal agreement, the pipeline recorded a formal agreement in force in 81 of 99 resolved cases. Where it reads a letter, the pipeline recorded one in 3 of 54. The two classifications concord on 86 percent of resolved documents. The residual runs in the direction one would expect, since a document can be a formal agreement and still have lapsed.

Most verdicts survive attack. The adversary refutes 3 existence claims of the 188 it could rule on, so 98.4 percent survive. It refutes 5 type claims of 189, so 97.4 percent survive. Survival differs sharply by cell. Every one of the 147 cases where I hold an instrument and the vendor records none survives, against 92.7 percent in the cell where both sources record nothing. This is again the asymmetry that establishing absence is harder than establishing presence.

Requiring all three passes to agree, 144 of the 147 cases behind the vendor comparison (98.0 percent) are triple-confirmed, as are 35 of 42 (83.3 percent) in the both-record-nothing cell. Two findings from the replication cut against me, and I report them as measured rates. Of 168 matched documents, 5 (3.0 percent) are not employment instruments, and 1 (0.6 percent) names a different person.

What the replication establishes is this. The anchoring defect was not load-bearing. The type split is not an artifact of the classifier's own labels. And the existence claims behind Section 5 do not fall to a rater whose instructions are to break them.

C A defect in the shipped adapter

The pre-correction target renderer emitted a bare `NaN` token for missing strings, and the model learned it. Of 1,992 fine-tuned outputs, 1,792 contain `NaN`. Python's permissive JSON parser accepts that token and a strict RFC 8259 parser does not, so strict parsing succeeds on only 194 outputs, against 1,985 for the untuned base. Downstream code parses permissively, so the dataset is unaffected. But the released artifact needs either a documented post-processing step or a retrain. I report the defect because a held-out F_1 cannot see it, and because it illustrates the prompt-versioning point of Section 2.2.

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